

Components of Computer (Software)

Topics

1. What is software?
2. Categories of software: System, Application software & Software Development Tools
3. Type and Functions of System Software (OS, Device Drivers & Utility Software)
4. Capabilities and Limitations of Computers
5. Relationship between hardware and software

Definition of Modern Computer

- A computer is a general-purpose device that can be programmed to carry out a set of arithmetic or logical operations automatically.
- It is a programmable electronic device that processes, stores, and manipulates data to perform a wide range of tasks and calculations.
- Computers are used in various fields and applications, including business, communication, education, entertainment, research, and more.
- The actual machinery or physical part of the computer (e.g. keyboard, screen display) — is called **HARDWARE**.
- The instructions and data are called **SOFTWARE**.

What is a Software?

Software means computer instructions or data & it includes anything that can be stored electronically - *webopedia.com*.

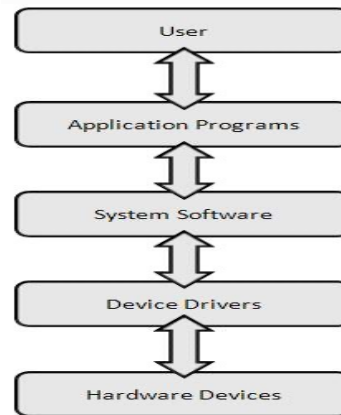
Software is a general term for the various kinds of programs used to operate computers and related devices - *techtargget.com*.

Software is any set of instructions that directs a computer to perform specific operations - *wikipedia.org*

Categories of Software

1. **System software** – operate and control the computer hardware, and to provide a platform for running application software.
 - Operating system – Windows, Linux, Mac
 - System utilities (OS helpers)
 - Device drivers - enable devices to function.
 - Utility Software
2. **Application Software** -- are tools that help users to perform tasks.
3. **Software Development Tools** – are used by programmers/ developers to develop other software (system or apps).

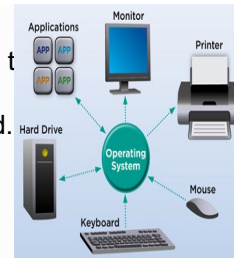
Simple Layered Interaction Model



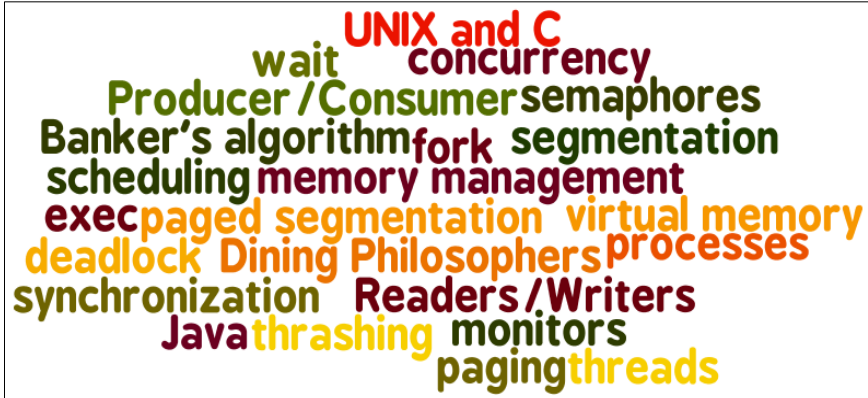
<http://cs.sru.edu/~mullins/cpsc100book/images/softwareLayers.jpg>

A. System Software

1. **Operating Systems (OS)** - manages hardware resources and provides a platform for other software to run.
 - Examples: Windows, macOS, Linux, and Android.
2. **Device Drivers** - facilitates communication between hardware devices and the operating system.
 - programs that enable a device/ hardware (e.g. printer, switches) to communicate with the computer
 - Included with the operating system or bundled with peripherals/ device.
3. **Utilities** -- Tools and programs that help manage and maintain the computer system, such as disk cleanup tools, antivirus software, and backup utilities



Functions of the Operating System



Source: <http://perugini.cps.udayton.edu/teaching/courses/Spring2014/cps346/>

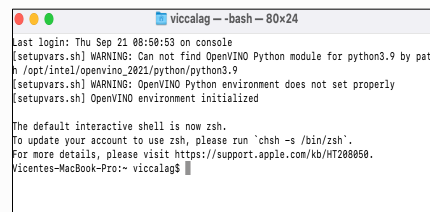
Functions of the Operating System

1. Interface between the user and the hardware

- graphical user interface (GUI)
- command-line interface (CLI) in which users type commands at the command-line interface (CLI) to tell the OS to do things.



GUI – Graphical User Interface

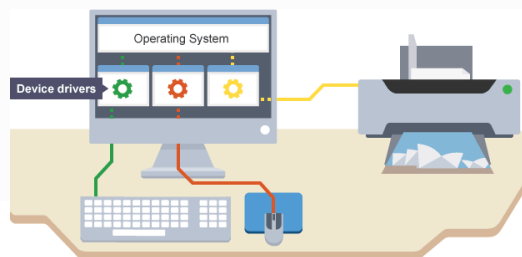


CLI – Command Line Interface

Functions of the Operating System

2. Coordinate hardware components

- enables coordination of hardware components.
- Each hardware device speaks a different language, but the operating system can talk to them through the specific translational softwares called device drivers.
- Every hardware component has different drivers for Operating systems. These drivers make the communication successful between the other softwares and the hardware.



Functions of the Operating System

3. Provide environment for software to function

4. File System Management

- **File Creation, Access, and Deletion** -- handles file operations, such as creating, reading, writing, and deleting files, and organizes them in a hierarchical structure.
- **File Permissions** -- enforces access control and permissions to protect files and data from unauthorized access.

5. Resource Monitoring and Management

- tracks CPU, memory, and disk usage to optimize resource allocation.
- We can see how busy our CPU is, or how quickly our hard drives retrieve data, or how much data our network card is sending etc. and it also monitors system activity for malware.
- It distributes workloads across multiple processors or resources for performance optimization (**Load Balancing**).
- Modern OSs include power-saving features to reduce energy consumption when the system is idle.

Resource Monitoring and Management

Activity Monitor								
All Processes								
CPU Memory Energy Disk Network								
Search								
Process Name	% CPU	CPU Time	Threads	Idle Wake Ups	% GPU	GPU Time	PID	User
Activity Monitor	24.2	2:21	12	11	0.0	0.00	3087	viccalag
WindowServer	13.4	23:20.15	14	29	0.0	7:57.91	170	_windowserver
Google Chrome Helper (Rend...	9.4	1:30.48	14	55	0.0	0.00	1147	viccalag
kernel_task	7.3	13:47.14	195	845	0.0	0.00	0	root
Google Chrome Helper (Rend...	5.2	58.25	14	53	0.0	0.00	2891	viccalag
Microsoft PowerPoint	4.2	8:38.80	43	84	0.0	13.12	526	viccalag
sysmond	2.6	4.33	5	0	0.0	0.00	403	root
TouchBarServer	2.4	1:09.61	8	12	0.0	12.21	389	root
VTDecoderXPCService	2.2	19.47	8	1	0.0	3.27	2230	viccalag
Microsoft Excel	2.1	4:02.64	27	13	0.0	0.91	516	viccalag
Google Chrome Helper (GPU)	2.1	8:39.73	12	63	0.0	3:46.73	1071	viccalag
com.apple.AppleUserHIDRiv...	1.6	1:25.63	4	0	0.0	0.00	368	_driverkit
launchservicesd	1.4	20.73	6	0	0.0	0.00	139	root
Epson Scanner Monitor	1.4	2:08.43	4	1	0.0	0.00	730	viccalag
Google Chrome	1.1	11:22.26	44	6	0.0	0.01	499	viccalag
tcdd	1.0	12.09	3	0	0.0	0.00	172	root
Google Chrome Helper	0.9	3:34.96	12	19	0.0	0.00	1072	viccalag
distnoted	0.8	38.01	2	0	0.0	0.00	446	viccalag

System: 9.40%
User: 13.91%
Idle: 76.69%

CPU LOAD

Threads: 3,232
Processes: 606

Jan 6, 2010

Functions of the Operating System

6. Process Management

- **Process Creation and Termination** -- creates, manages, & terminates processes (programs in execution) to enable multitasking and efficient resource utilization.
- **Scheduling** -- determines which processes are allowed to run, when they run, and for how long, ensuring fairness and efficiency in CPU resource allocation.

7. Memory Management:

- **Memory Allocation** -- assigns and deallocates memory to processes, ensuring that they have the memory they need for execution.
- **Virtual Memory** -- manages virtual memory spaces, allowing the OS to use disk storage as an extension of physical RAM.

8. Device Management

- **Device Drivers** -- communicates with hardware devices through device drivers, enabling input and output (I/O) operations and device control.
- **I/O Operations** -- manages data transfers between devices and memory, ensuring data integrity and synchronization.

Functions of the Operating System

9. Security and Access Control

- **User Authentication** - manages user accounts, passwords, and access privileges to control user access to the system and resources.

10. Backup and Recovery

- **Data Backup** -- backup utilities to create copies of files and system configurations.
- **System Recovery** -- allows for system restoration in case of system failures or disasters.

Characteristics of Operating System

1. Licensing

- **Open Source OS** – free to download and modify
 - Linux - Ubuntu, Debian, Fedora, CentOS, openSUSE, Linux Mint, Arch Linux, elementary OS, Manjaro, Zorin OS
 - OpenBSD -- OpenBSD is a security-focused Unix-like operating system known for its emphasis on code correctness, security auditing, and robust cryptography.
 - NetBSD - NetBSD is another Unix-like operating system known for its portability, supporting a wide range of hardware platforms
- **Free OS** – free and not open source (you cannot modify them)
 - Example: Chrome OS and makes it free to use.
- **Commercial OS** – not free, can be used with payment
 - Examples: Microsoft Windows and Apple macOS

Characteristics of Operating System

2. Software Compatibility

- The developers make the softwares which may be compatible or incompatible in different versions within the same operating system's type but they can't be compatible with the other OS types.
- Every OS type have their own software compatibility.

3. Complexity

- Operating systems come in basically two editions one is 32-bit and other is 64-bit editions.
- The 64-bit edition of an operating system best utilizes random access memory (RAM).
- A computer with a 64-bit CPU can run either a 32-bit or a 64-bit OS, but a computer with a 32-bit CPU can run only a 32-bit OS.

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System Software: Device Drivers



<http://study.com/academy/lesson/systems-software-utility-software-device-drivers-firmware-gui.html>



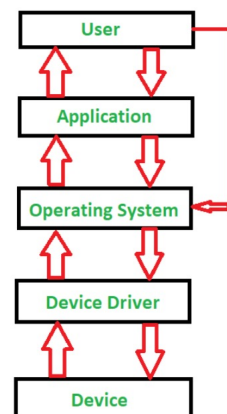
<https://medium.com/>

Functions of Device Drivers

Device drivers -- facilitate communication between the operating system (OS) and hardware devices.

Device drivers are necessary to permit a computer to interface and interact with specific devices.

They define the messages and mechanism whereby the computer -- the OS and applications -- can access the device or make requests for the device to fulfill.



Functions of Device Drivers

1. **User Interface Integration** -- allow users to adjust device settings, update firmware, or troubleshoot issues related to the hardware.
2. **Device Initialization** -- initializes the device, configuring it to a state where it can be used by the OS and applications
3. **Data Transfer and I/O Control** -- facilitate the transfer of data between the CPU and the hardware device. They manage input and output operations (I/O) by providing functions or methods that applications can call to read from or write to the device.
4. **Error Handling and Reporting** -- detect errors or malfunctions and report them to the operating system, allowing for error recovery and diagnostic information that can be logged or presented to the user.
5. **Compatibility and Optimization.** Device drivers are often updated to improve compatibility with new hardware, enhance performance, and fix bugs. Manufacturers release updated drivers to ensure that their devices work seamlessly with the latest operating systems and applications.
6. **Plug-and-Play Support** -- devices can be connected or removed while the system is running.

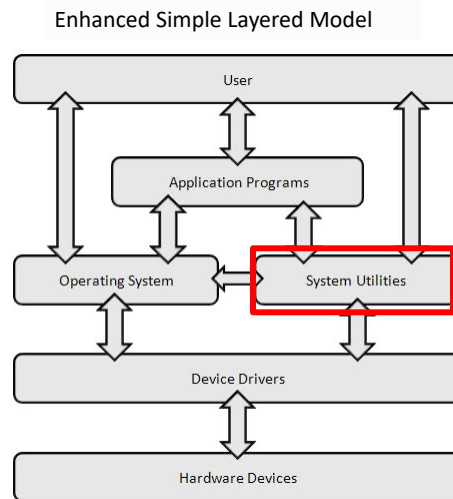
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System Software: Utility Software

- highly specialized and designed to perform only a single task or a small range of tasks.
- a system software designed to help analyze, configure, optimize and maintain the computer.
- Although utilities are part of the system software, they are not part of the OS per se. They are loaded into memory as needed, either by the user or the OS.



Source: <http://cs.sru.edu/~mullins/>

Functions of Utility Software

- a. Serve as tools for system maintenance and repairs that aren't automatically handled by the operating system
- b. Make it easier for users to:
 - Copy files between storage devices
 - Repair damaged data files
 - Translate files so that different programs can read them
 - Guard against viruses and other potentially harmful programs
 - Compress files so they take up less disk space
 - Perform other important tasks

Functions of Utility Software

1. Disk Cleanup and Optimization -- scan and clean up unnecessary files, temporary files, cache, and log files to free up disk space and improve system performance.
2. Disk Defragmentation -- rearrange fragmented files on a hard drive to optimize data storage and retrieval, leading to faster file access times.
3. System Maintenance -- automate routine maintenance tasks such as updating software, checking for system errors, and repairing corrupted files.
4. Backup and Recovery - create backups of important files, folders, and system configurations. It also offers tools for recovering lost or accidentally deleted data.

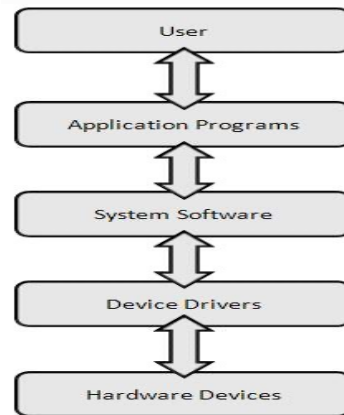
Functions of Utility Software

5. Security and Antivirus - safeguard the system and data from online and offline threats, helping to maintain system integrity and privacy.
6. Password Managers -- store and manages passwords for various online accounts, simplifying login and enhancing security.
7. File Compression and Archiving -- compress files and folders to reduce storage space and create archive files (e.g., ZIP or RAR) for easy sharing.
8. Network Tools
9. Uninstaller Programs
10. Privacy and Cleanup Tools
11. Data Encryption and Security

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Simple Layered Interaction Model



<http://cs.sru.edu/~mullins/cpsc100book/images/softwareLayers.jpg>

Application Software

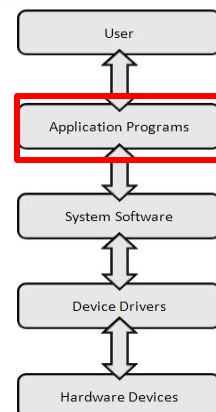
An application software or app is computer software designed to help the user to perform a specific task.

The system software serves the application while the application software serves the user.

Two general categories of application software:

- horizontal applications
- vertical applications

Simple Layered Interaction Model



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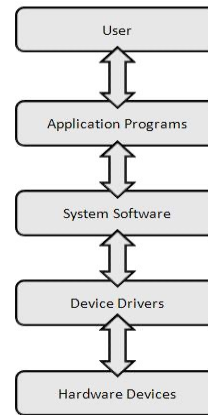
Application Software

System & Application software:

Examples:

1. Productivity tools – Word Processing, Electronic Spreadsheets, Presentation, Database Management systems
2. Accounting software, graphics software, communication software, game software, etc
3. Entertainment software – games, video
4. Statistical software – SPSS, SAS, etc
5. GIS & Remote Sensing software – ArcGIS, mapinfo, Idrisi

Simple Layered
Interaction Model



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Software Licenses

1. Commercial / Retail /Proprietary Software
 - a computer software that is produced for sale.
2. Pre-installed software
 - trial basis
 - purchased for you by the computer manufacturer (OEM or Original Equipment Manufacturer).
3. Shareware Software
 - normally provided with restricted usability, either with a time limit or without all of the features.
 - Shareware is copyrighted and licensed.

Software Licenses

4. Freeware

- It is copyrighted, licensed software that is available for free or for an "optional cost", i.e., a donation. There is lots of freeware on *tu cows* also.

5. Free Open Source Software (FOSS)

- software that is free and
- It comes with the source code that users may modify and redistribute.

Categories of Software

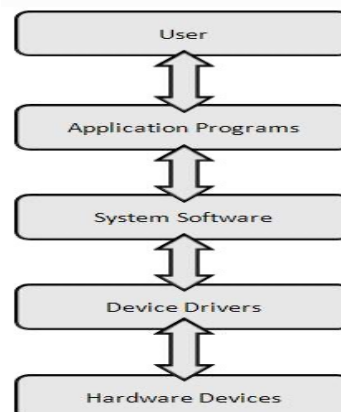
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Software Development Tools

Programming tools are programs or applications that software developers (also known as *programmers*, *coders*, *hackers* or *software engineers*) use to create, debug, and maintain software.

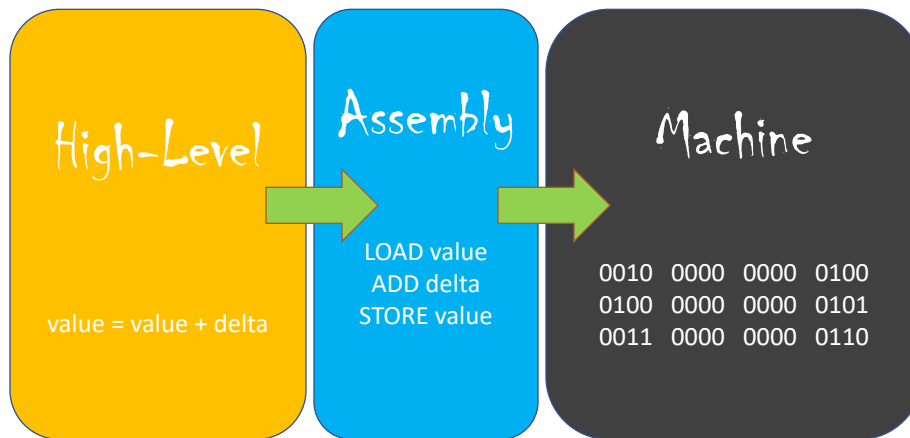
These tools may be relatively self-contained programs such as compilers, debuggers, interpreters, linkers, and text editors, that can be combined together to accomplish a task; or they may form an integrated development environment (IDE), which combines much or all of the functionality of such self-contained tools.

Programming Languages

✓ Categories of Computer Language

1. Machine Language -- numeric codes that represent data (0, 1)
2. Assembly language (add eax, ebx ; adds ebx to eax)
3. High-level language -- falls between machine language and natural human language
 - Compilers translate high-level language into machine language
4. Natural Languages -- resemble languages spoken by humans

Programming Languages



The Language of Computers

Machine and Assembly languages can be used only on the computer where it was programmed.

High-level languages are more popular than machine and assembly languages.

High-level languages can be used in any computer even without modification on the program.

Other Software - Malicious Software

Malware (Malicious Software)

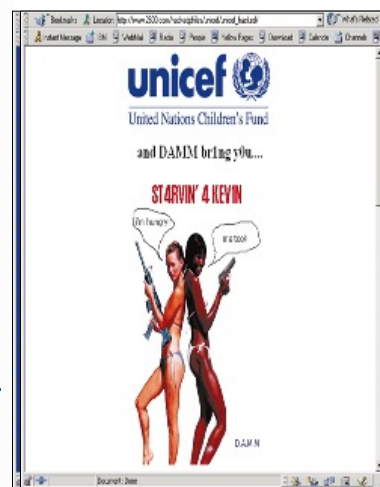
It consists of programming (code, scripts, active content, and other software) designed to:

1. *disrupt or deny operation,*
2. *gather information that leads to loss of privacy or exploitation,*
3. *gain unauthorized access to system resources, and*
4. *other abusive behavior.*

Other Software - Malicious Software

Purpose of Malware:

1. Pranks - intended to be harmless or merely annoying, rather than to cause serious damage to computer systems.
2. Intentionally Harmful - programs are designed to cause harm or data loss (e.g. Explozezip), vandalize web pages (e.g. [Code Red](#) worm in the 2001)



Other Software - Malicious Software

Purpose of Malware:

3. Profit
4. Hacktivism
5. Virus and worms
6. Trojans or Trojan horse -- is any program that invites the user to run it, concealing a harmful or malicious payload. The payload may take effect immediately and can lead to many undesirable effects, such as deleting the user's files or further installing malicious or undesirable software.
7. Spyware - is a type of malicious software that can be installed on computers, and which collects small pieces of information about users without their knowledge.

Other Software - Malicious Software

Purpose of Malware:

8. Keystroke logging - is the action of tracking (or logging) the keys struck on a keyboard, typically in a covert manner so that the person using the keyboard is unaware that their actions are being monitored.

Key logging is often used by law enforcement, parents, and jealous or suspicious spouses (lovers).

References:

1. What is a computer? www.webopedia.com
2. <http://perugini.cps.udayton.edu/teaching/courses/Spring2014/cps346/>
3. <http://cs.sru.edu/~mullins/cpsc100book/images/softwareLayers.jpg>
4. <https://medium.com/computing-technology-with-it-fundamentals/operating-system-its-functions-and-characteristics-c0946e4215c6>

Thank you!